Advanced Time Series Project

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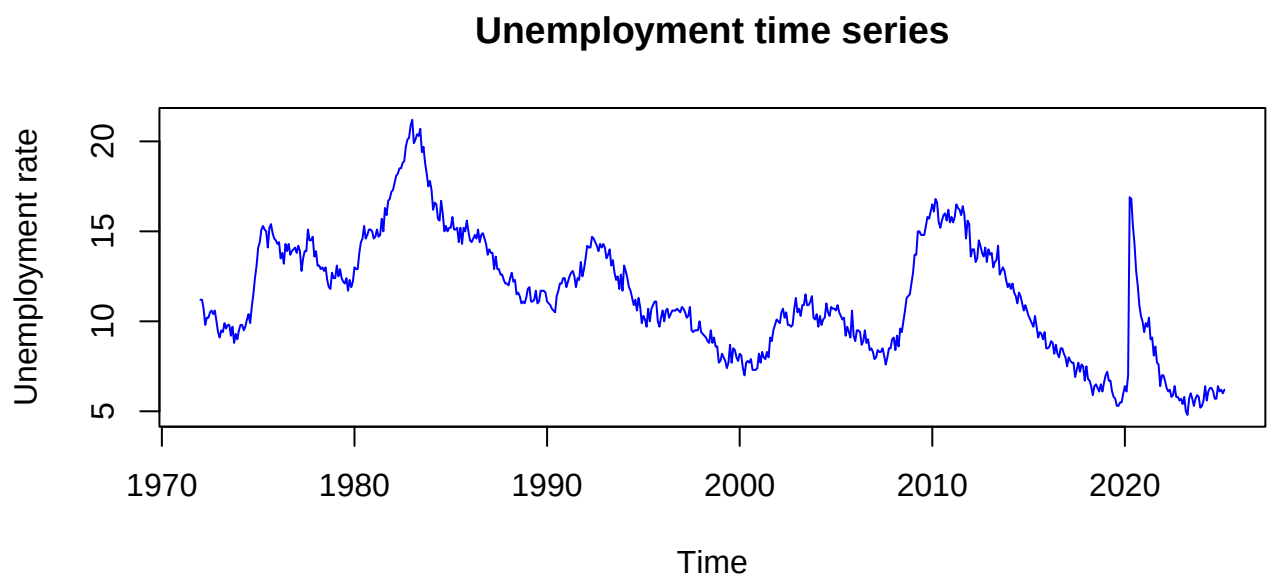
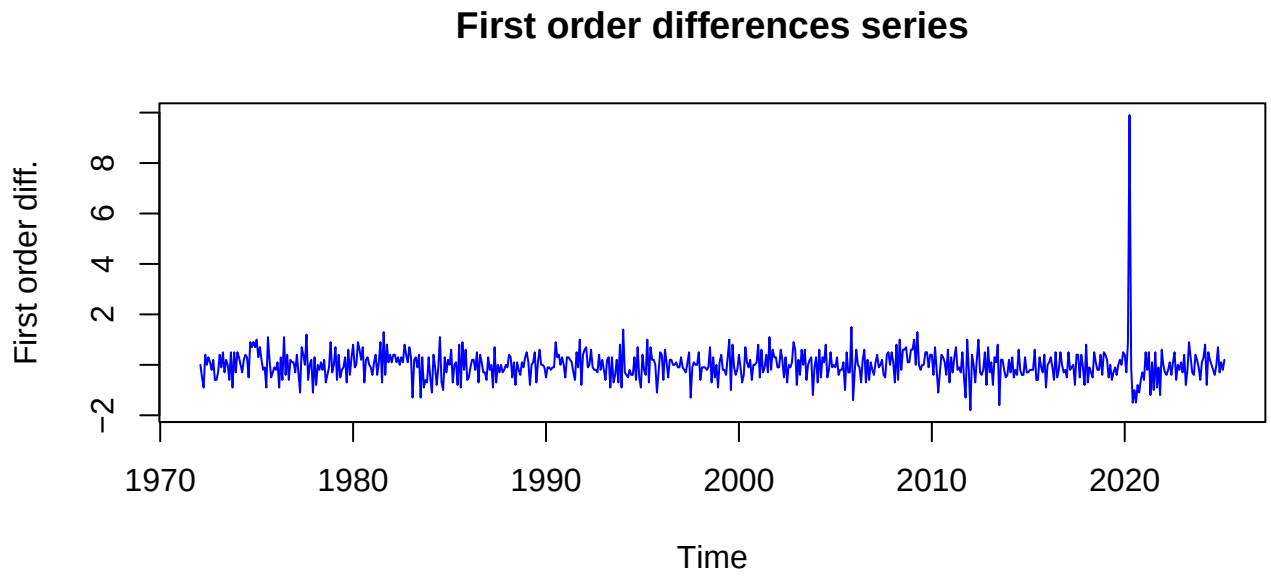
The data

The data for this analysis come from the U.S. Bureau of Economic Analysis, Unemployement rate, retrieved from FRED, at https://fred.stlouisfed.org/series/PSAVERT, April 29, 2025. Once loaded the useful packages (stats and tseries), the first thing to do was to plot the time series.

```
df <- read.csv("LNS14000006.csv")
ts_data <- ts(df$LNS14000006, start=c(1972,1), frequency=12)

# Plot the time series and its difference
## time series
ts.plot(ts_data,xlab="Time",ylab="Unemployment rate",
        main="Unemployment time series",col="blue")
```




```
## first order differences
ts.plot(diff(ts_data),xlab="Time",ylab="First order diff.",
        main="First order differences series",col="blue")
```



At first glance, it is clear that the *unemployment rate* was affected by the last two greatest crisis: the 2008 financial crisis and the COVID-19 one from 2020, even though the second one was more impactful in the short term (as it can be seen from the plot of the first order differences of the time series).

First approach: classical time series decomposition through moving averages smoothing

Since the variance seems to remain stable regardless the value of the series, we can try and decompose the series with an additive model. The data are already seasonally adjusted, therefore what remains to obtain is the deterministic component.

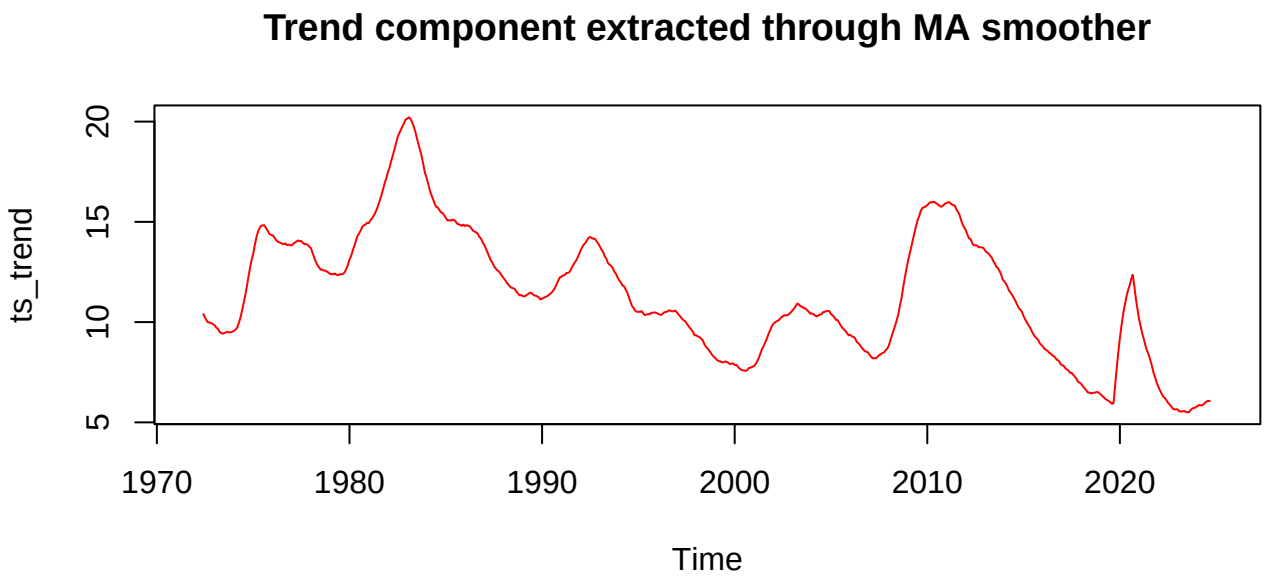
```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

ts_trend <- stats::filter(ts_data, rep(1/12, 12), sides = 2)
#smoothing the series with a finite Moving Average filter

ts_residual <- ts_data - ts_trend # should be a stationary random noise component
ts_residual <- na.omit(ts_residual)
mean(ts_residual) # the residuals shows null mean
```

```
## [1] 0.003410297
```

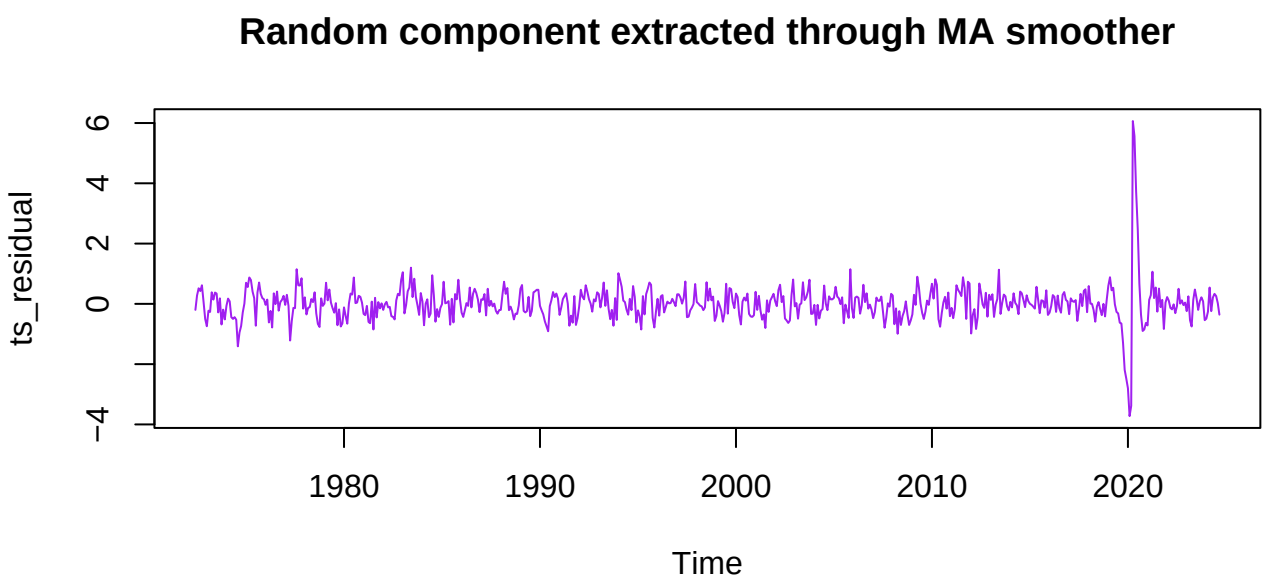
```r
adf.test(ts_residual)
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ts_residual
## Dickey-Fuller = -10.917, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

```r
#the Augmented Dickey Fuller test shows that this series is indeed stationary

ts.plot(ts_trend, col= "red",
        main = "Trend component extracted through MA smoother")
```




```r
ts.plot(ts_residual, col = "purple",
        main = "Random component extracted through MA smoother")
```



Thus, we extracted the trend component and the noise component from the time series.

Second approach: analysis with linear models for time-varying parameter

While classical decomposition through moving averages provides a useful baseline by extracting a deterministic trend, macroeconomic variables like the unemployment rate are frequently subject to unpredictable structural shocks. To properly account for this dynamic nature, a stochastic trend approach is required. Therefore, we transitioned to a time-varying parameter framework, specifically exploring the Local Level Model (LLM)

We firstly checked if the series was stationary by using Augmented Dickey Fuller Test for stationariety

```
adf.test(ts_data, alternative = "stationary")
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ts_data
## Dickey-Fuller = -2.8975, Lag order = 8, p-value = 0.1984
## alternative hypothesis: stationary
```

Since the p-value is 0.1984, the series is not stationary. Therefore, we tried to differentiate it until it, eventually, became stationary.

```
diff_ts_data <- diff(ts_data)
adf.test(diff_ts_data, alternative = "stationary")
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  diff_ts_data
## Dickey-Fuller = -8.5544, Lag order = 8, p-value = 0.01
## alternative hypothesis: stationary
```

The first order differences seemed to be stationary given that the p-value of the Augmented Dickey Fuller Test is equal to 0.01.

SF1 unconditional moments, ACF and PACF of the time series

Then, we moved to compute the SF1 unconditional moments of the time series.

```
mean.series <- mean(ts_data)
var.series <- var(ts_data)
skewness.series <- mean(((ts_data-mean.series)/(var.series^0.5))^3)
kurtosis.series <- mean((ts_data-mean.series)^4)/(var.series^2)
```

```
## mean: 11.37418
```

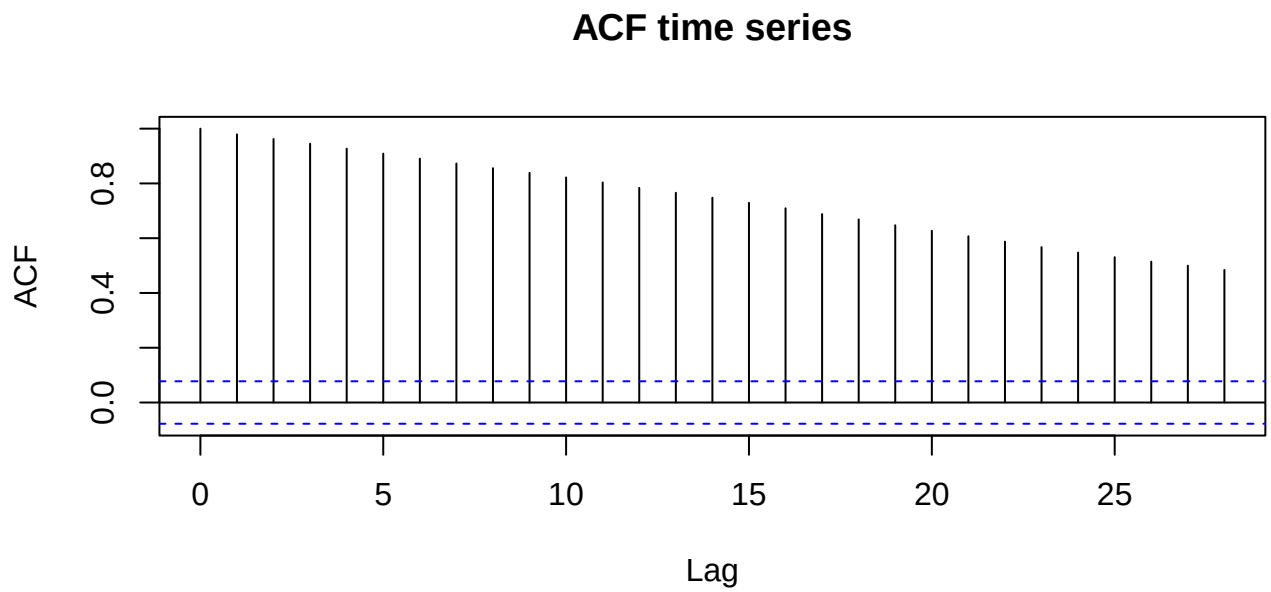
```
## variance: 11.02881
```

```
## skewness: 0.2822598
```

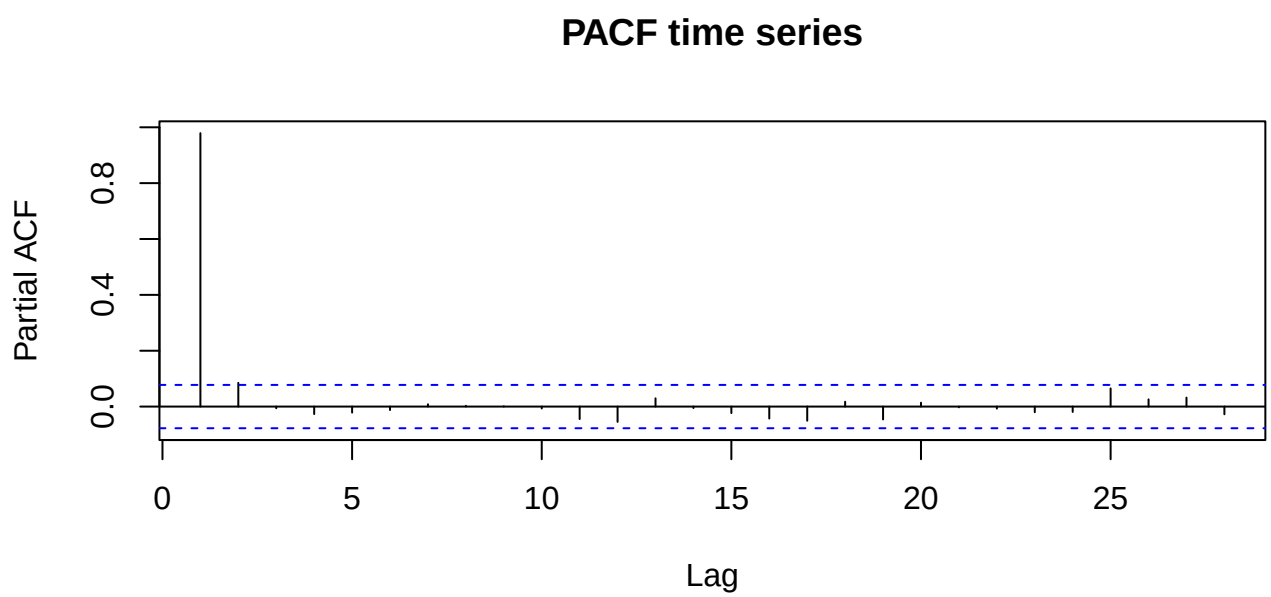
```
## kurtosis: 2.674078
```

Obtained the unconditional moments it remained to see the behaviour of the autocorrelation (ACF) and partial autocorrelation function (PACF).

```
acf(as.vector(ts_data),main="ACF time series")
```




```
pacf(as.vector(ts_data),main="PACF time series")
```

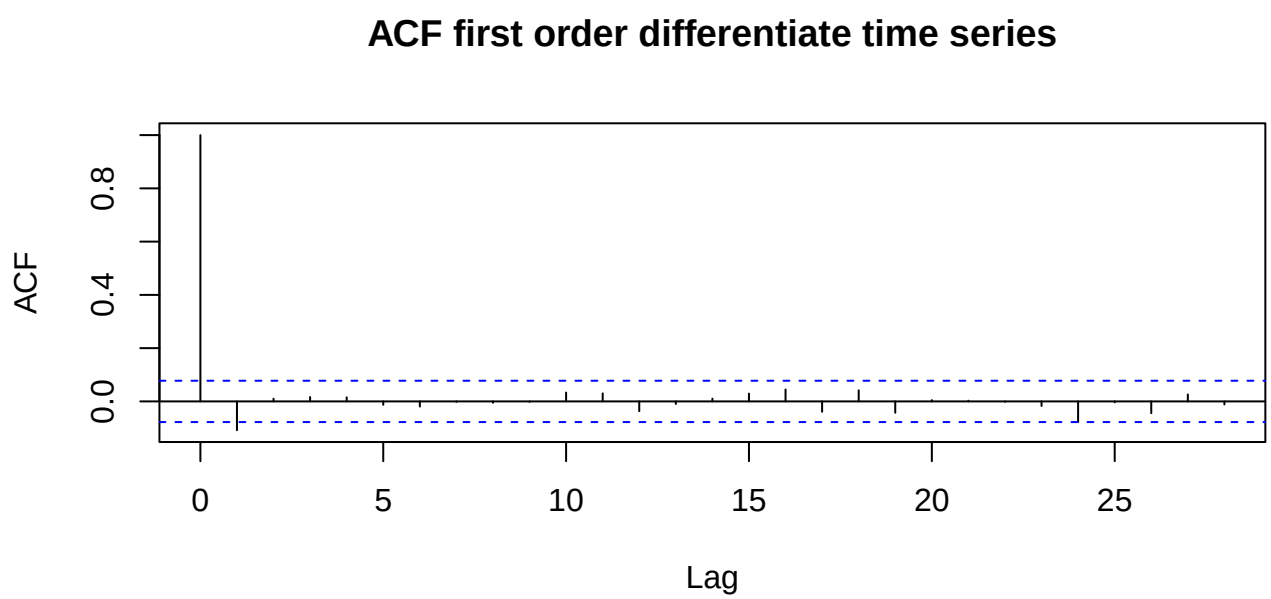


The slow, almost linear decay of the ACF, paired with a highly significant peak at lag 1 in the

PACF, is the textbook signature of a non-stationary, integrated process of order 1, is commonly referred to as a Random Walk.

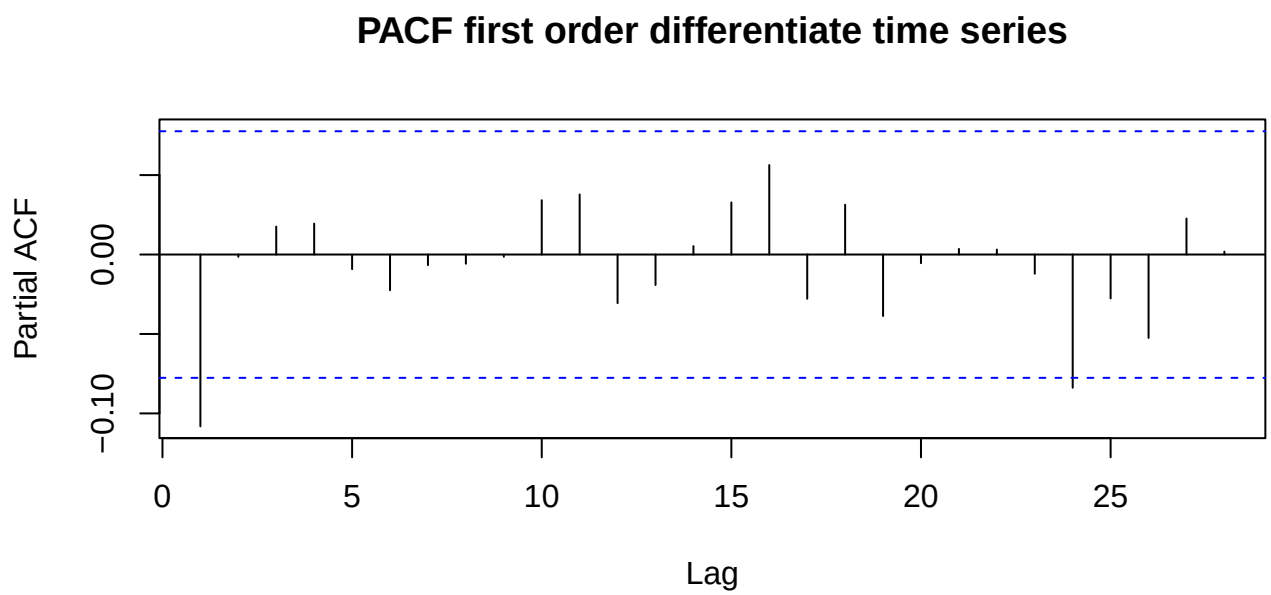
This visually confirms the results of the ADF test and justifies the need to difference the series. Indeed, the ACF and the PACF of the once differentiated time series, is stationary, looked like an MA(1), exactly as the theory of an LLM (Local Level Model) suggests.

```
acf(as.vector(diff_ts_data),main="ACF first order differentiate time series")
```



After lag 2 the fluctuations of the ACF do not exceed the significance bounds,that suggest that differencing has made the series stationary.

```
pacf(as.vector(diff_ts_data),main="PACF first order differentiate time series")
```



The ACF cuts off after lag 1, without any significant spikes at subsequent lags. This behaviour strongly supports an MA(1) structure for the differenced series, which perfectly aligns with the theoretical reduced form of a Local Level Model

Local Linear Model (LLM)

A Local Level Model (LLM) is a model of the type:

$$y_t = \mu_t + \epsilon_t,$$

where:

- $\mu_{t+1} = \mu_t + \eta_t$, $\eta_t \sim NID(0, \sigma_\eta^2)$,
- $\epsilon_t \sim NID(0, \sigma_\epsilon^2)$,
- $\epsilon_t \perp \eta_t \ \forall t, s$,

The reduced form of Δy_t=$y_t - y_{t-1}$ is:

$$\Delta y_t = \eta_{t-1} + \epsilon_t - \epsilon_{t-1},$$

or, from another formulation:

$$\Delta y_t = \xi_t - \theta \xi_{t-1}, \quad \xi_t \sim NID(0, \sigma_\xi^2),$$

The model

We estimated the value of θ for the MA(1) representation of Δy_t.

```
fit_arima <- arima(ts_data,order=c(0,1,1))
theta <- fit_arima$coef
sigma2_xi <- fit_arima$sigma2
```

We estimated a value of θ for -0.1062, and a value for σ_ξ^2 of 0.4141. Theoretically, the value of θ should be included between 0 and 1 but, given our formulation, the bounds to consider are -1 and 0. Since we know that $\sigma_\xi^2 = \frac{\sigma_\eta^2}{(1+\theta^2)}$, we simply obtained σ_η^2 by solving the equation above for it, obtaining a value of 0.4206.

```
sigma2_eta <- sigma2_xi*(1 + theta)^2
```

Lastly, the variance of the error term, ϵ_t, is obtainable by the formula of the autocovariance at lag 1 (γ_1) of Δy, since the LLM is a MA(1), and so $\sigma_\epsilon^2 = -\theta * \sigma_\xi^2$.

```
sigma2_epsilon <- -theta*sigma2_xi
```

We obtained a value of 0.044 for σ_ϵ^2 and 0.331 for σ_η^2. Since $\sigma_\epsilon^2 > 0$, the underlying process is not a pure random walk, but rather a random walk plus noise. The signal-to-noise ratio $q = \frac{\sigma_\eta^2}{\sigma_\epsilon^2}$ is approximately 7.5. This indicates that the variance of the signal dominates the variance of the noise, which suggests that the Kalman filter will be able to track the observed series quite closely.

Forecasting with the Kalman filter

Inital conditions

To initialize the Kalman filter algorithm, it is mandatory to specify the starting conditions, namely $\mu_{1|0}$ and $P_{1|0}$. In this specific case, we opted for $\mu_{1|0} = y_1$. For the initial variance of the state prediction error, we set $P_{1|0} = \sigma_\epsilon^2$, assuming the error in our first state estimate is simply the measurement error. Then, $\mu_{t+1|t}$, $P_{t+1|t}$ and all the other values (v, K, F) were calculated iteratively.

```r
n <- length(ts_data)
y <- as.numeric(ts_data)

q <- sigma2_eta/sigma2_epsilon # signal to noise ratio

mu_pred <- numeric(n)  # mu_{t|t-1}
P <- numeric(n)
v <- numeric(n)
K <- numeric(n)
F <- numeric(n)
llk <- numeric(n)

mu_pred[1] <- y[1]
P[1] <- sigma2_epsilon
```

The SSF for a Local Level Model Any linear time series model can be cast into a State Space Form, defined by a measurement equation and a transition equation:

$$y_t = Z_t\alpha_t + \epsilon_t\alpha_{t+1} = T_t\alpha_t + \eta_t$$

Specifically for our Local Level Model, the matrices and unobserved components collapse into scalars as follows

- $Z_t = 1$ (representation matrix),
- $\alpha_t = \mu_t$,
- $G_t = \sigma_\epsilon$ (scale matrix),
- $T_t = 1$ (transition matrix),
- $H_t = \sigma_\eta$,
- $Q_t = 1$,

The iteration

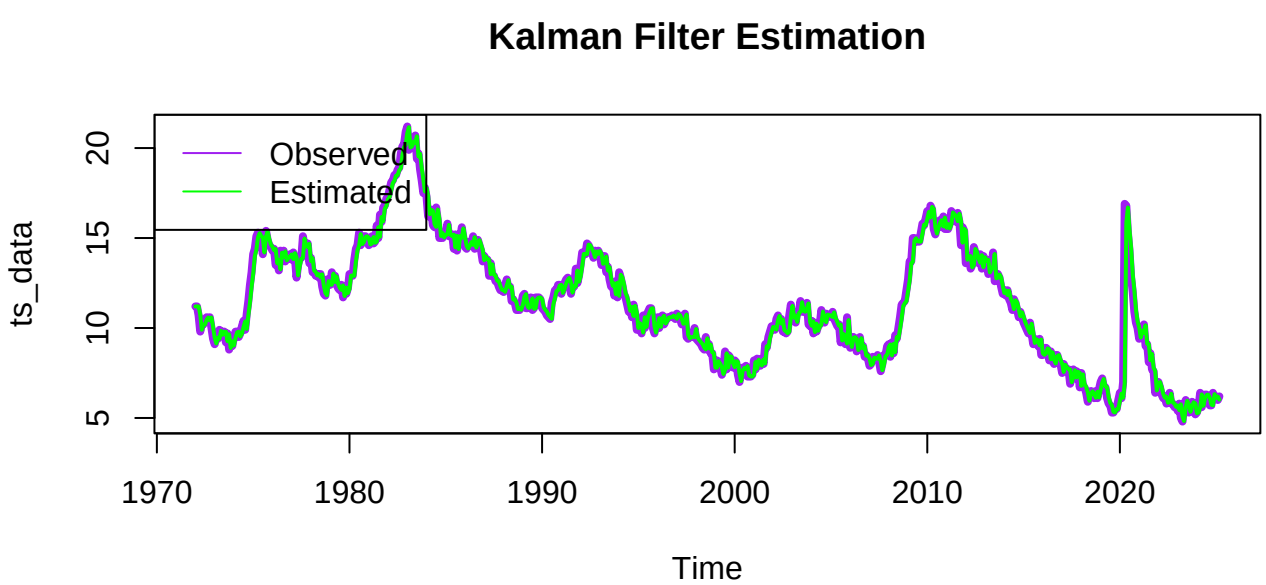
The Kalman filter algorithm iteratively calculate the prediction of some quantities at time t+1 given the information at time t. It simply works as follows:

- given a prediction $y_{t|t-1}$, once the time t arrives, y_t is observed and the prediction error $\upsilon_t = y_t - y_{t|t-1}$ is computed;
- then, the innovation variance $F_t = P_t + \sigma_\epsilon^2$ is calculated;
- the Kalman gain is computed as $K_t = \frac{P_t}{F_t}$;
- the new prediction for y_{t+1} is $y_{t+1} = y_t + K_t\upsilon_t$;
- and the variance of the prediction, $P_{t+1|t} = P_t + \sigma_\eta^2 - K_tF_tK_t$.

```r
for (t in 1:(n-1)) {
  # Prediction error
  v[t]<-y[t]-mu_pred[t]
  # Innovation variance
  F[t]<-P[t]+sigma2_epsilon
  # Kalman gain
  K[t]<-P[t]/F[t]
  # Update step
  mu_pred[t+1]<-mu_pred[t]+K[t]*v[t]
  P[t+1]<-P[t]+sigma2_eta-K[t]*F[t]*K[t]
  # Log-likelihood
  llk[t]<--0.5+log(F[t]+v[t]^2/F[t])
}

loglik <- sum(llk)
```

Finally, we plotted the time series and the results from the Kalman filter.

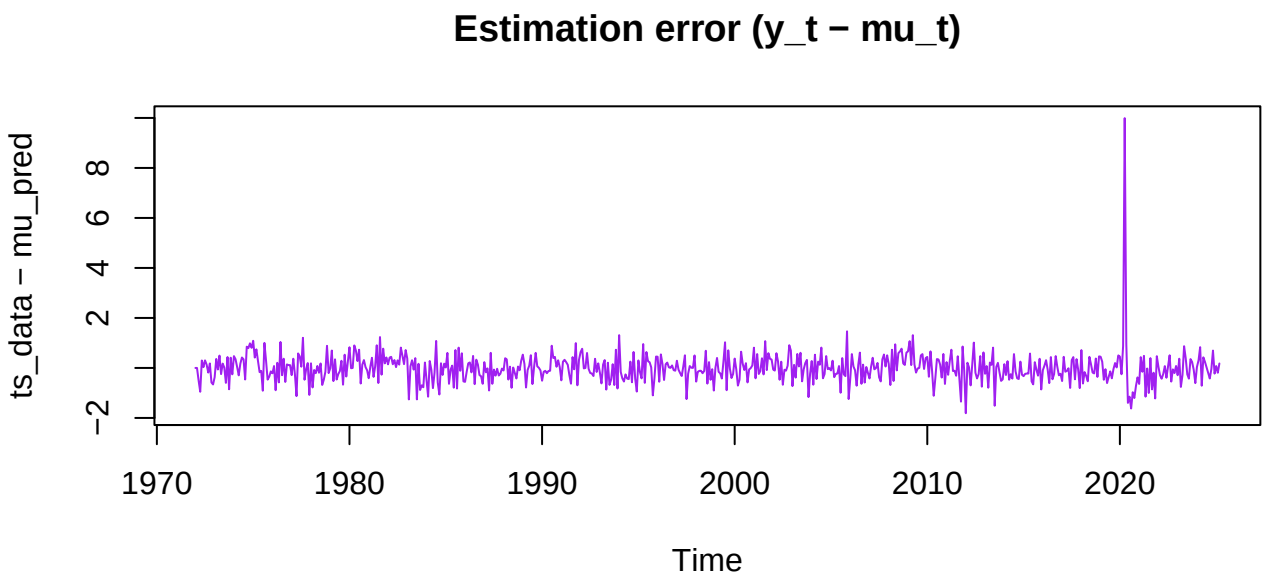
```r
ts.plot(ts_data, main = "Kalman Filter Estimation",
        col = "purple",type = "l", lwd = 4)
lines(ts(mu_pred, start = start(ts_data), frequency = frequency(ts_data)),
      col = "green",lwd=2)
legend("topleft", legend = c("Observed", "Estimated"),
       col = c("purple", "green"), lty = 1)
```



The Kalman filter captures the signal very effectively, as evidenced by the two curves largely overlapping. Because the estimated signal-to-noise ratio (≈ 7.5) is relatively high, the model assigns substantial weight to the new observations, adapting quickly to the structural changes in

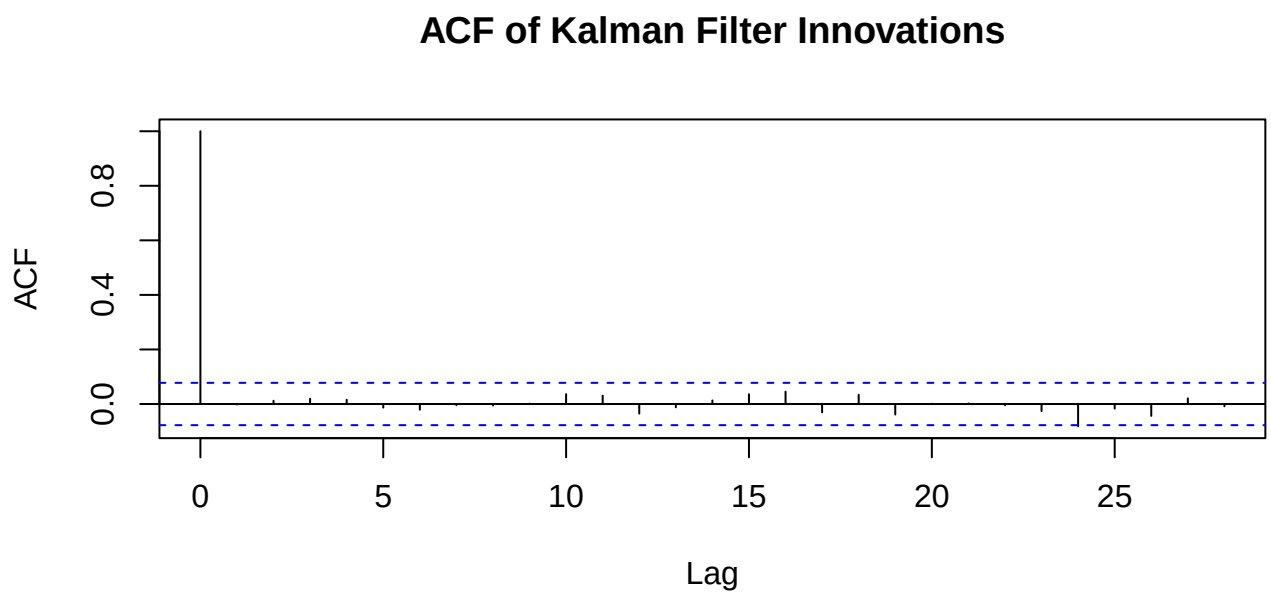
the series while still applying a slight smoothing effect due to the presence of the noise variance.

```
ts.plot(ts_data - mu_pred, col = "purple", main = "Estimation error (y_t - mu_t)")
```



The estimation error fluctuates around zero, indicating that overestimations and underestimations are generally balanced. At certain points the peaks significantly exceed the confidence bounds, suggesting that while the model adapts well overall, it inevitably lags slightly behind abrupt, steep changes in the data.

```
innovations <- v[-1]
acf(innovations, main="ACF of Kalman Filter Innovations")
```




```r
Box.test(innovations, lag = 12, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  innovations
## X-squared = 3.2166, df = 12, p-value = 0.9938
```

To definitively validate our State Space Model, we must ensure that the Kalman filter has captured all the systematic dynamics of the time series. If the model is correctly specified, the prediction errors, or innovations (v_t), should be purely random, resembling a White Noise process.

By plotting the ACF of the innovations v_t we can observe whether any significant autocorrelation remains. Furthermore, the Ljung-Box test can formally confirm the null hypothesis of no autocorrelation. If the ACF shows no significant spikes and the test yields a high p-value, we can confidently conclude that our Local Level Model is adequate and the Kalman filter extracted all the available signal from the data.

The role of q: signal-to-noise ratio

The quantity q, called the signal-to-noise ratio, is defined as the ratio between the signal variance σ_η^2 and the noise variance σ_ϵ^2:

$$q = \frac{\sigma_\eta^2}{\sigma_\epsilon^2}.$$

If q tends to zero, the noise variance dominates the signal variance; conversely, if q diverges, the signal variance dominates the noise variance. The value of q affects the ability of the Kalman Filter to capture the underlying signal. When q is small, the Kalman Filter assigns more weight to the predictive model rather than to the observations.

The Kalman Filter and the parameter q are linked through the matrix P, since the Kalman Gain at time t is defined as:

$$K_t = \frac{P_t}{F_t} = \frac{P_t}{P_t + \sigma_\epsilon^2},$$

and the evolution of the prediction error variance is given by:

$$P_{t+1|t} = P_t + \sigma_\eta^2 - K_t^2 F_t.$$

Intuitively, the Kalman Gain acts as a weighting mechanism. If the measurement noise variance is large relative to the uncertainty of our prediction, K_t becomes small, and the filter relies heavily on its own internal model, resulting in a smoother trend. Conversely, in our specific dataset where the signal dominates (high q), the Kalman Gain remains high. This forces the filter to highly trust the incoming observations, adapting instantly to sudden economic shifts like the 2008 and 2020 crises. We need an initial value for P, which is typically set as:

$$P_1 = q = \frac{\sigma_\eta^2}{\sigma_\epsilon^2}.$$

Now let's investigate how the Kalman Filter empirically behaves as a function of q, by defining hypothetical values for the two variances that are not related to our time series:

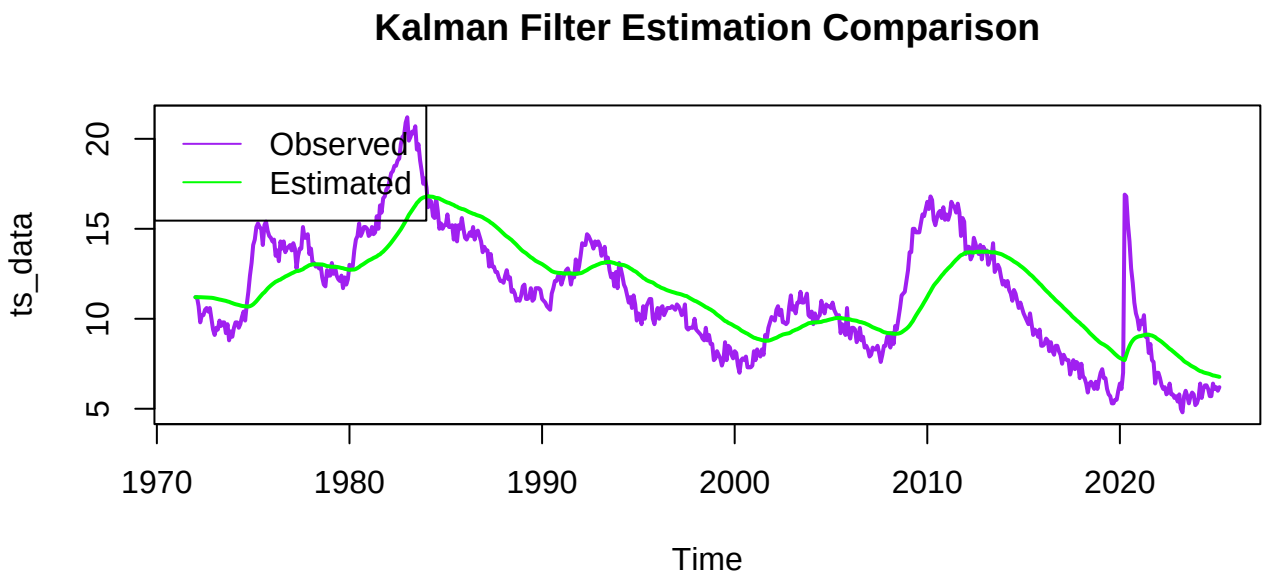
When q has a smaller value than what has been estimate:

```r
# New initial conditions
ts_data <- ts(df$LNS14000006, start = c(1972, 1), frequency = 12)
n <- length(ts_data)
y <- as.numeric(ts_data)
sigma2_eta_com1 <- 0.01 # new variances
sigma2_epsilon_com1 <- 10

q <- sigma2_eta_com1/sigma2_epsilon_com1 # signal to noise ratio comparison

mu_pred <- numeric(n)  # mu_{t|t-1} comparison
P <- numeric(n)
```

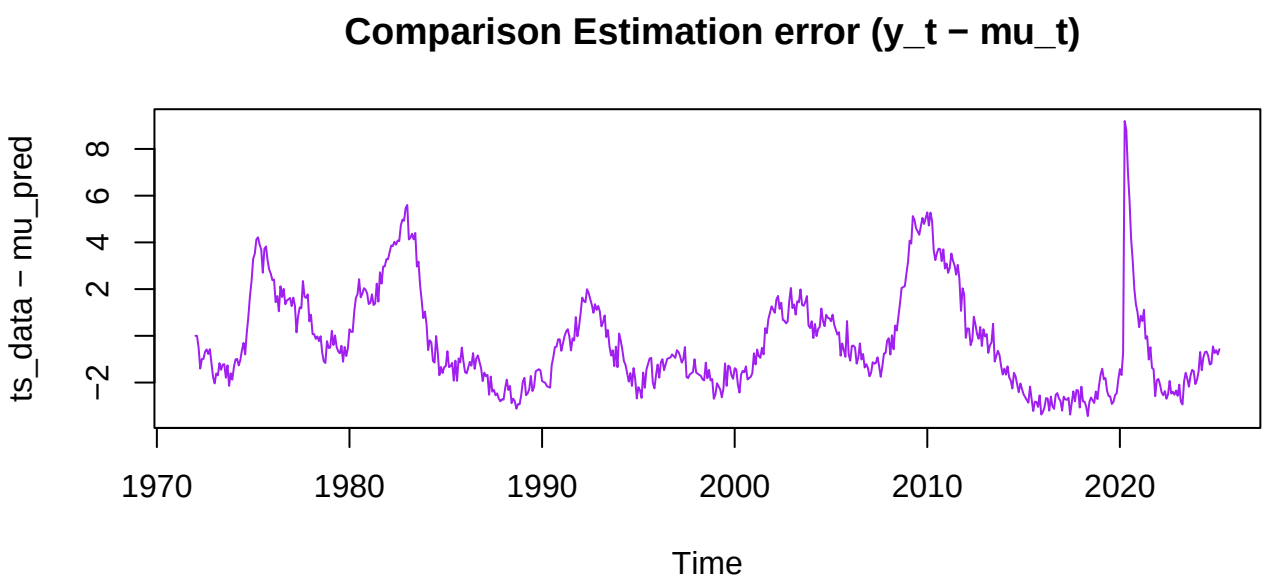
```r
v <- numeric(n)
K <- numeric(n)
F <- numeric(n)
llk <- numeric(n)

mu_pred[1] <- y[1]
P[1] <- q

# Iteration
for (t in 1:(n-1)) {
  # Prediction error
  v[t]<-y[t]-mu_pred[t]
  # Innovation variance
  F[t]<-P[t]+sigma2_epsilon_com1
  # Kalman gain
  K[t]<-P[t]/F[t]
  # Update step
  mu_pred[t+1]<-mu_pred[t]+K[t]*v[t]
  P[t+1]<-P[t]+sigma2_eta_com1-K[t]*F[t]*K[t]
  # Log-likelihood
  llk[t]<--0.5+log(F[t]+v[t]^2/F[t])
}

loglik <- sum(llk)
```

Now let's see the plots:

```r
ts.plot(ts_data, main = "Kalman Filter Estimation Comparison",
        col = "purple",type = "l", lwd = 2)
lines(ts(mu_pred, start = start(ts_data), frequency = frequency(ts_data)),
      col = "green",lwd=2)
legend("topleft", legend = c("Observed", "Estimated"),
       col = c("purple", "green"), lty = 1)
```




```
ts.plot(ts_data - mu_pred, col = "purple",
        main = " Comparison Estimation error (y_t - mu_t)")
```



As expected, by drastically lowering the value of q, the Kalman Filter behaves in a much more rigid and smoothed manner. It treats the massive spikes (such as the 2008 and 2020 shocks) primarily as random noise rather than a structural shift in the underlying state. Consequently, the estimated trend fails to capture these severe fluctuations, leading to massive spikes in the estimation error plot compared to our initial, correctly specified model.